Lab 6: Identifying Data for Regression

Exploring possible outcomes and predictors for Report 2

FALL 2026 · WEEK 08 — WEEK OF MONDAY 2026-10-12, IN SECTION

DELIVERABLES — DUE AT THE END OF YOUR LAB SECTION, ON BRIGHTSPACE

- ☐ `Lab-6_FIRSTNAME_LASTNAME.Rmd/Lab-6_FIRSTNAME_LASTNAME.qmd`: the starter sheet, renamed, with the four parts completed as far as you got and the generative AI disclosure filled in
- ☐ `Lab-6_FIRSTNAME_LASTNAME.pdf`: the same sheet, knitted to PDF
- ☐ the data file the sheet reads, and any image files you add

Report 2 uses multiple linear regression to examine how a numeric outcome relates to several predictors considered together. The outcome is what we want to explain or predict, while the predictors are other measurements or group characteristics that may help account for its variation.

You can reuse your Report 1 dataset or find another one. Please check what the possible outcomes measure and examine their distributions before exploring relationships with other variables. These findings provide a basis for choosing an outcome and predictors that suit your question. Your choices here are provisional, so please review them against its brief when it is released.

What to download and where to put it

Two files are on Brightspace:

- this handout;
- the starter sheet, `lab-06-identifying-data-for-regression.Rmd` *or* `lab-06-identifying-data-for-regression.qmd`.

Please make a folder for the lab inside your course folder, put the starter in it, and put your data in a `data/` folder beside it. Open the starter in RStudio or VSCode and put your name in its header. Please rename the file: `Lab-6_FIRSTNAME_LASTNAME.Rmd` *or* `Lab-6_FIRSTNAME_LASTNAME.qmd`.

Lab 2's Part 1, "Finding a dataset", lists places to look for a file. Its source rules apply: no dataset that ships with R, has been used in this course, is generally used for data science examples, or comes from the UC Irvine Machine Learning Repository. Data from another course you have taken or are taking are also excluded. Research data you have permission to analyze in this course are allowed. There is no list of approved datasets. If you are unsure, please send the dataset link to satadisha.saha.bhowmick@vanderbilt.edu.

If you have no usable file today, please ask your TA for help. You can practice on `PluckData.csv` from the earlier lectures, as in Lab 3. It ***cannot*** be used for the report. Please record the need for a report dataset in Part 4.

What to hand in and when

Please submit the renamed Rmd/qmd, its knitted PDF, and the files it reads on your lecture section's Brightspace page before you leave. The syllabus allows generative AI here with disclosure of the tool and purpose (or "none" if unused). Please check its output and be ready to explain your work.

> **NOTE — How this lab is graded**
> The syllabus rubric applies. Labs are out of 5 points: one for arriving before the section starts, three for progress (at least half of the sheet by the end of the section), and one for the TA checking your progress when you hand it in. Half of this worksheet is Parts 1 and 2. If you miss your section, please get the lab from your TA and complete all four parts within one week of your original lab time for the three progress points. The two attendance points are not available, and attending another section is not an option under the syllabus.

> **HINT — If knitting to PDF will not work**
> Please use the syllabus's fallback: knit to HTML, convert that to PDF with an online converter such as `https://www.sejda.com/html-to-pdf`, or your browser (as shown in class) and submit the PDF and Rmd/qmd. Please also tell your TA.

How the two hours are divided

The times include writing your answers as you work.

Activity	Minutes
Read the handout and open the files	10
Part 1: data and possible outcomes	25
Part 2: distributions	30
Part 3: relationships	30
Part 4: provisional choice and next checks	20
Knit, show your TA, and submit	5

Part 1: Data and possible outcomes

A numeric column can contain a measurement, a count, a rating, a category code, or an identifier. Please use the documentation to decide what it means. Its storage type in R/Python does not establish that.

The data and the question

Please record the source and documentation, collector and purpose, unit of observation, and license. You can adapt these fields from your Report 1 dataset card. Check whether the data were observed or generated, and prefer observed data as in Lab 2.

Look for a numeric outcome, several predictors, and enough observed rows. Please identify repeated measurements of the same subject, since these observations can be dependent.

> **TASK 1 — Open and describe the data**
> Please load your data and use `dim()`, `names()`, `glimpse()`, and `summary()` or corresponding Python equivalents. Record changes and check the row count again. Write a question about how a numeric outcome relates to other measurements.

What could be an outcome?

Please examine two possible outcomes, even if you already prefer one. Use the sheet's table to record each one's meaning, scale, units, documented and observed ranges, distinct values, and missing values.

An interval or ratio measurement has meaningful differences between values, such as ten dollars or one centimeter. A customer number does not. Seven distinct values, or seventy, do not establish a measurement scale.

Binary outcomes belong to the later classification work. Ordered ratings and counts concentrated at zero need a different scale judgment from an ordinary measurement. If only one outcome is suitable, compare it with a numeric-looking variable you rule out. If neither works, please record why and ask your TA for help finding another file.

> **TASK 2 — Check the measurements and compute summaries**
> Please complete the sheet's variable table and compute the mean, median, standard deviation, minimum, and maximum for plausible outcomes. Check documented missing codes first, such as 999, and record any recoding. A real extreme value should not be recoded as missing.

The integers 1 to 10 might be ordered satisfaction categories. A measurement rounded to centimeters is still a measurement even though its recorded values are integers. A predictor can be categorical even when the outcome is numeric. Please keep useful group variables for Part 3.

Part 2: Distributions

Please draw a histogram and a boxplot for each plausible outcome. Describe its center, spread, asymmetry, gaps, and any values that need checking. Try a second histogram bin width if the shape depends on the first one. Please label the axes with the variable and its units.

The observed range and the possible range

Compare the figures with the possible values from the documentation. A sample containing only high-income households has a restricted range of income, even if every recorded income is correct. A rating with most answers at its upper boundary has a ceiling effect. Please say which observations are absent or uncommon and how that limits the comparison you can make.

A point beyond a boxplot whisker is a value to inspect. It might be a recording error, a documented missing code, or a real observation. Please check before changing it, and do not remove a value just to make the distribution look more symmetric.

> **TASK 3 — Draw and interpret the distributions**
>
> Please make the figures in Part 2 of the sheet and write a paragraph for each outcome. Include its mean, median, and range from Part 1, using inline R or Python where possible. Say what you would keep, check, or change and why. If you propose a transformation, say what problem it might address and whether zero or negative values would prevent using a logarithm. You do not need to transform a variable today.

Normality and the outcome

An outcome does not have to have a normal distribution to be used in linear regression. The usual normality assumption for its tests and intervals concerns the model's errors: the differences between the outcomes and their expected values given the predictors. After a model is fitted, we examine residuals (observed outcomes minus fitted values) to assess this assumption. A histogram of the outcome alone cannot do that. The NIST explanation sets out why the error distribution matters for inference.

Please describe skew and unusual values now, then revisit them when a model has been fitted. This lab does not require a normality test or a regression fit. A large skewness or kurtosis statistic is a reason to examine a distribution, not a universal pass/fail rule.

Optionally, try `shapiro.test()` or the Python equivalent `shapiro()` from `scipy.stats` module and the sheet's question about power. The test assumes a normal distribution under its null hypothesis. Its R help page specifies 3 to 5,000 nonmissing values. A small p value is evidence against normality, and a large one does not prove it. Neither decides whether to keep an outcome. With more data, a small departure can be detected more easily.

Part 3: Relationships

Please choose two plausible predictors for each suitable outcome. Before computing a correlation, explain why the measurements might relate. Consider how and when they were collected. A value calculated from the outcome itself, such as a total that includes it, can give a strong relationship without an informative explanation or prediction. For a prediction question, the predictor must also be available at the time the prediction is needed.

A figure before a correlation

For a numeric predictor, please draw a scatterplot with the predictor on the horizontal axis and the outcome on the vertical axis. For a categorical predictor, please draw a boxplot of the outcome in each group and record the group sizes. Describe any direction, curve, change in spread, separate groups, or unusual observations.

Pair	A useful summary
Two measured quantities	Pearson's r describes linear association. Spearman's ρ describes association in their ranks, including a curved relationship that consistently increases or decreases.
Numeric outcome and ordered categories	Spearman or Kendall can describe an ordered relationship. Check the order of the codes in the documentation.
Numeric outcome and two groups	Pearson's r with the groups coded 0 and 1 is the point-biserial correlation. Name which group is 1.
Numeric outcome and three or more unordered groups	Compare the group distributions and summaries. A correlation of arbitrary numeric group codes depends on how you number the groups.

Pearson's r can be computed without normal distributions but its significance testing and interval have assumptions beyond that computation. Please choose a summary according to the measurement scale and the relationship you see, rather than using a normality test to select a method. Spearman's ρ can describe a monotonic curve (one that consistently rises or falls), but neither r nor ρ summarizes a U-shaped relationship well.

Missing values and interpretation

Please create a table containing just the outcome and predictor for each calculation, then keep the rows observed on both. Record how many rows remain. `cor()` returns `NA` by default if a contributing value is missing. The R help page explains the missing-value options and the rank methods. If you compare Pearson and Spearman, use the same rows for both. A variable with no variation cannot have a defined correlation. Please report a correlation's sign and size, with its row count. There is no minimum $|r|$ that qualifies a predictor for this lab. A weak bivariate relationship does not establish that a variable would add nothing when other predictors are included. A

strong one does not establish a cause or reliable prediction on new observations.

> **TASK 4 — Explore and explain the relationships**
> Please complete the plots, calculations, and relationship table in Part 3. Interpret each pair using the figure as well as the statistic. No p values or significance screening are required.

Other explanations for a relationship

Please choose one relationship and name a third variable that might help explain it, using the confounding examples from the correlation lecture. Say why that variable might affect both the proposed predictor and the outcome, and whether it was measured. A diagram is optional. A smaller association after adjustment would not by itself prove that the causal explanation is right. Please record the question here; no partial or semi-partial correlation calculation is required.

Part 4: A provisional choice

Please choose the outcome you would investigate first and two predictors, or explain why you need a different dataset. Compare your outcome with the alternative you examined, using its meaning, the usable rows, the figures, and your question. The most symmetric variable or the largest correlation is not automatically the best choice for a question you care about.

> **TASK 5 — Write the plan**
> Please complete Part 4 of the sheet: the question, outcome and units, possible predictors and the reason for each, the alternative you considered, two checks still needed, and what your present analysis cannot establish. Please say if you used the practice dataset and still need a file for the report.

> **NOTE — Nothing chosen today is binding**
> This is practice in evaluating data for regression. You can change the dataset, question, outcome, or predictors as you learn more. Please use the Report 2 brief when it is released to check your plan. Finishing the sheet does not mean that a regression model's assumptions have been checked. Those depend on the model you fit.

Before you submit

Please restart R and knit the sheet from its saved folder. Check that the figures, tables, and written answers appear in the PDF. Submit the Rmd/qmd, PDF, and files it reads, with the disclosure completed, and show your TA your progress before you leave.